



CAPSTONE PROJECT

TAT ANALYSIS: OPERATIONAL EFFICIENCY POC

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OUTLINE:

- **Problem Statement**
- **Proposed System/Solution**
- **System Development Approach**
- **Algorithm & Deployment**
- **Result (Output Image)**
- **Conclusion**
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PROBLEM STATEMENT:

- The organization wants to analyze Turn Around Time (TAT) performance across laboratories, departments, tests, and geographies.
- There is a need to identify operational bottlenecks, measure SLA adherence, and improve efficiency by tracking whether tests are completed within defined TAT thresholds.
- Stakeholders require a clear understanding of overall TAT performance, variations across different dimensions, and sample journey delays.

PROPOSED SOLUTION:

- **Dashboard Development:** Development of Power BI dashboards for comprehensive TAT analysis.
- **KPI Tracking:** Calculation of Key Performance Indicators (KPIs) based on the provided TAT dataset.
- **Comparative Insights:** Enabling comparative analysis across various dimensions, including labs and departments.
- **Journey Mapping:** Implementation of hourly sample journey analysis to identify peak load hours and delay-prone processing stages.

SYSTEM APPROACH:

- **Platform:** The project utilizes Power BI for POC-level data modeling and visualization.
- **Data Sources:** Data is integrated from TAT Transaction Data (registration, processing, and completion timestamps) and Master Data (Lab, Department, Test, City, State).
- **Architecture:** The data model is designed as a Star schema featuring single-direction filtering.
- **Tables:** It consists of a Fact Table for Test Transactions and Dimension Tables for Lab, Department, Test, Date, and Geography.

ALGORITHM & DEPLOYMENT:

- **Scope:** Predictive analytics and machine learning algorithms are explicitly out of scope for this POC.
- **Core Logic:** The primary calculation algorithm focuses on deriving metric percentages, such as calculating % In TAT as $(\# \text{ Tests In TAT} / \# \text{ Total Tests Performed}) * 100$.
- **Deployment Delivery:** The final solution is deployed and delivered as a Power BI .pbix file.
- **Interactivity:** Visuals are configured to interact dynamically with slicers for Date Range, Lab, Department, Test Type, State, and City.

RESULT:



Summary Page

KPI Names

Tests in TAT

KPIs

Visuals

Detailed Page

The top 29 City all had City % IN TAT of 100.00%. Across all 240 City, City % IN TAT ranged from 35.14% to 100.00%.

At 95.89%, IMMUNOASSAYS had the highest % IN TAT and was 902.87% higher than GENERAL EXAMINATION, which had the lowest % IN TAT at 9.56%.

Across all 18 Department Name, % IN TAT ranged from 9.56% to 95.89%.

At 31122, 12:00 - 13:00 had the highest Samples Dept to Result and was 21,512.50% higher than 03:00 - 04:00, which had the lowest Samples Dept to Result at 144. Samples Dept to Result and total Samples Reg to SRA are positively correlated with each other. Across all 25 SRA Hour Slab, Samples Dept to Result ranged from 144 to 31122, Samples Reg to SRA ranged from 144 to 31326, and Samples Result to Approval ranged from 144 to 31047.

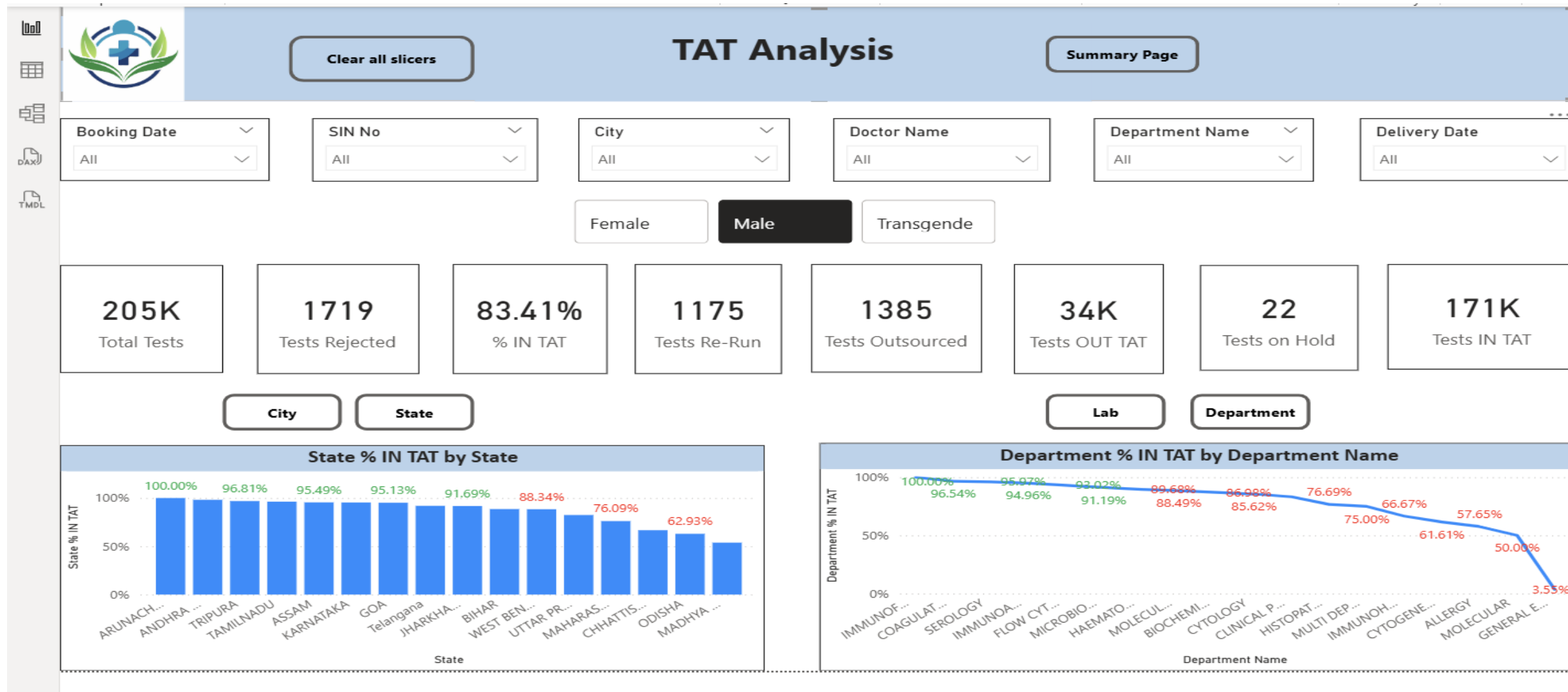
PUDUCHERRY and ARUNACHAL PRADESH tied for highest State % IN TAT at 100.00%, followed by KARNATAKA. MADHYA PRADESH had the lowest State % IN TAT at 81.56%. Across all 17 State, State % IN TAT ranged from 81.56% to 100.00%.

413.15K

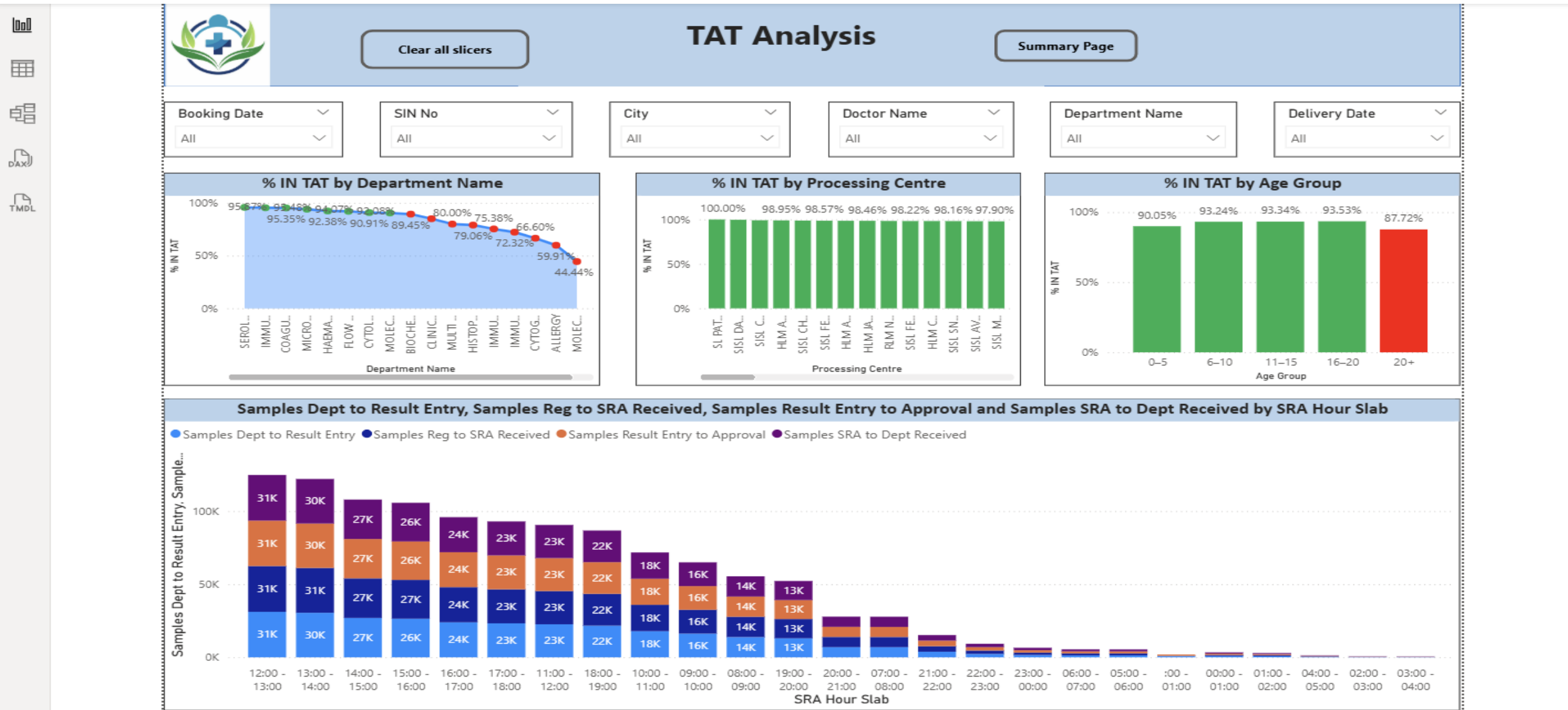
KPI Value

For More Detailed Information about Health - [World Health Organization](#)

RESULT:



RESULT:



RESULT:



Clear all slicers

Detailed Page

Summary Page

Booking Date

All

SIN No

All

City

All

Doctor Name

All

Department Name

All

Delivery Date

All

Test Name	State	Visit No	SIN No	Sex	Processing Centre	Actual Test Centre	Age	Approval Status	Booking Centre
IMMUNOGLOBULIN IGM	ANDHRA PRADESH	LMRT18734	BI04407553	Male	RRL VIJAYAWADA	NRL MUMBAI	70 Y 0 M 0 D	Y	RRL VIJAYAWADA
IMMUNOGLOBULIN IGM	ASSAM	LTLR263	BI04277487	Female	RRL GUWAHATI	NRL MUMBAI	43 Y 0 M 0 D	Y	PCC RANGIA 1
IMMUNOGLOBULIN IGM	CHHATTISGARH	LRAI55541	BI04417790	Female	SL RAIPUR	NRL MUMBAI	70 Y 0 M 0 D	Y	SL RAIPUR
IMMUNOGLOBULIN IGM	MADHYA PRADESH	LRAQ38461	BI04334215	Male	PCL JABALPUR	NRL MUMBAI	49 Y 0 M 0 D	Y	PCL JABALPUR
IMMUNOGLOBULIN IGM	MAHARASHTRA	L26L575	BI04338181	Male	PCL JALGAON	NRL MUMBAI	52 Y 0 M 0 D	Y	SIS CITYCARE DIAGNOSTICS
IMMUNOGLOBULIN IGM	MAHARASHTRA	LA9F3006	LD01664744	Female	SL PUNE	NRL MUMBAI	67 Y 0 M 0 D	Y	COCO BALEWADI
IMMUNOGLOBULIN IGM	MAHARASHTRA	LLOK15072	LD01334649	Male	SL ANDHERI	NRL MUMBAI	84 Y 0 M 0 D	Y	PCC LOKHANDWALA
IMMUNOGLOBULIN IGM	MAHARASHTRA	LNAS139784	BI04429681	Male	SL NASHIK	NRL MUMBAI	55 Y 0 M 0 D	Y	SL NASHIK
IMMUNOGLOBULIN IGM	MAHARASHTRA	LNAS139991	BI04435764	Male	SL NASHIK	NRL MUMBAI	64 Y 0 M 0 D	Y	SL NASHIK
IMMUNOGLOBULIN IGM	MAHARASHTRA	LSLO52161	BI04342736	Male	SL KOLHAPUR	NRL MUMBAI	78 Y 0 M 0 D	Y	SL KOLHAPUR
IMMUNOGLOBULIN IGM	WEST BENGAL	L9I5101	BI04271351	Female	HLM CAMRI	NRL MUMBAI	61 Y 0 M 0 D	Y	ASIAN CENTRE OF ONCOLOGY AND
IMMUNOGLOBULIN IGM	WEST BENGAL	L9I5112	BI04409921	Female	HLM CAMRI	NRL MUMBAI	65 Y 0 M 0 D	Y	ASIAN CENTRE OF ONCOLOGY AND
IMMUNOGLOBULIN IGM	WEST BENGAL	LBRT3620	BI04434502	Female	RRL KOLKATA	NRL MUMBAI	68 Y 11 M 7 D	Y	PCC TOBIN ROAD
IMMUNOGLOBULIN IGM	WEST BENGAL	LHYC1749	BI04299843	Male	RRL KOLKATA	NRL MUMBAI	64 Y 0 M 0 D	Y	COCO ASANSOL
IMMUNOGLOBULIN IGM	WEST BENGAL	LHYC1856	BI04401347	Female	RRL KOLKATA	NRL MUMBAI	53 Y 0 M 0 D	Y	COCO ASANSOL
IMMUNOGLOBULIN IGM	WEST BENGAL	LKOL258689	BI04413222	Male	RRL KOLKATA	NRL MUMBAI	71 Y 0 M 0 D	Y	RRL KOLKATA
IMMUNOGLOBULIN IGM	WEST BENGAL	LKOL260132	BI04431329	Male	RRL KOLKATA	NRL MUMBAI	71 Y 0 M 0 D	Y	RRL KOLKATA
IMMUNOGLOBULIN IGM	WEST BENGAL	LKOL260395	BI04435547	Female	RRL KOLKATA	NRL MUMBAI	78 Y 0 M 0 D	Y	RRL KOLKATA
1,3 BETA-D-GLUCAN	BIHAR	LMQH1124	SE00635049	Female	RRL PATNA	NRL MUMBAI	20 Y 0 M 0 D	Y	MEDIVERSAL HEALTHCARE PRIVATE
1,3 BETA-D-GLUCAN	BIHAR	LMQH1167	SE00640390	Male	RRL PATNA	NRL MUMBAI	72 Y 0 M 0 D	Y	MEDIVERSAL HEALTHCARE PRIVATE
1,3 BETA-D-GLUCAN	BIHAR	LPAT103047	SE00641837	Female	RRL PATNA	NRL MUMBAI	84 Y 0 M 0 D	Y	RRL PATNA
14Q32.3 BY FISH IGH GFNF	MAHARASHTRA	IRAM6626	CG00015123	Male	SI NAGPUR	NRI MUMBAI	65 Y 0 M 0 D	Y	COCO RAMDAS PFTH

CONCLUSION:

- The POC successfully enables data-driven operational improvements for the diagnostics business unit.
- Management can now efficiently monitor overall test volumes and TAT compliance.
- The business has gained crucial visibility into how test rejections, re-runs, holds, and outsourcing impact overall turnaround time.

FUTURE SCOPE:

- **Automation:** Transitioning the data refresh strategy from manual updates to a daily automated refresh for production environments.
- **Expansion:** Scaling the solution beyond POC-level data modeling to accommodate larger data volumes through advanced aggregations and optimization techniques.
- **Security:** Implementing user-level security (Row-Level Security), which was excluded from the initial POC.

REFERENCES:

- **Laboratory TAT Analytics:**

Cassim, N., et al. (2020). *Assessing the impact of a turn-around time dashboard at a high-volume laboratory*. African Journal of Laboratory Medicine. (This study demonstrates the real-world operational impact of TAT dashboards).

- **Data Modeling:**

Microsoft Learn. (n.d.). *Model relationships in Power BI Desktop*. (Official documentation outlining the Star Schema architecture and single-direction filtering utilized in this project).

- **GitHub Reference:**

<https://github.com/gopiashokan/Healthcare-Data-Analysis-using-PowerBI> (A public repository showcasing a comparable healthcare Power BI dashboard, including data modeling and DAX KPIs).

Thank You